

Stroop Effect Simulation using Drift Diffusion Model

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Introduction

The Stroop Effect, first introduced by John Ridley Stroop (1935), is one of the most well-known phenomena in cognitive psychology. It occurs when individuals are asked to name the ink color of a word while ignoring its semantic meaning. When the word and ink are incongruent (e.g., the word “red” printed in blue ink), response times increase and errors become more frequent.

This effect reflects the interference that occurs between competing cognitive processes, specifically between the automatic process of reading and the controlled process of color naming. Because of this, the Stroop task is widely used to study selective attention and cognitive control.

Although traditional studies rely on human participants, computational models offer an alternative method for studying these processes. In this study, a Drift Diffusion Model is used to simulate participant responses in both congruent and incongruent conditions, aiming to later analyze reaction times and error rates for all trials.

Research Objective

The objective of this experiment is to investigate the Stroop effect, by comparing the results of different simulated responses in congruent and incongruent trials. This allows us to gain insight into selective attention and cognitive control.

Literature Review

The Stroop Effect was first demonstrated by John Ridley Stroop in 1935, who found that individuals take longer to name ink colors when they conflict with the word meaning. The experiment has since been replicated multiple times and is considered a fundamental example of cognitive psychology.

Later Colin MacLeod (1991) published a comprehensive review of the Stroop Effect, highlighting that reading is an automatic process and therefore it is difficult to suppress even when it is irrelevant to the task.

To simulate decision-making processes Roger Ratcliff and McKoon (2008) introduced

the Drift Diffusion Model (DDM). This model is used to analyze both accuracy and reaction times in a decision.

Methodology

Instead of human participants, a Drift Diffusion Model was used to simulate reaction times and decision accuracy for both congruent and incongruent trials. The model parameters were designed to reflect varying levels of cognitive interference from conflicting word information.

Simulation Procedure:

Model: Drift Diffusion Model

Drift rate computation: I calculated the drift for each trial as:

$$\text{Drift} = (\text{color weight} * \text{color signal}) + (\text{word weight} * \text{word signal})$$

Congruent trials: color and word signals aligned (+1).

Incongruent trials: color and word signals conflicted (+1 vs -1).

Other Parameters:

Decision threshold: 0.8

Noise: 0.5

Time step: 0.001

Maximum trial duration: 5.0

I ran 500 trials per condition for congruent and incongruent trials. For each trial I used the `stroop_drift` function to compute the reaction time and error rate and varied the word weight to examine how increasing cognitive conflict affected the performance.

Results

The simulation produced clear results in both trials. In incongruent trials the results indicate that the reaction time and error rate are higher than in congruent trials. We can observe that as word weight increases, the mean reaction time sharply increases in incongruent trials. While in congruent trials an increase in word weight signifies a decrease in reaction time. Meanwhile error rates is near zero in congruent trials regardless of word weight. Lastly, in incongruent trials, error rate increases dramatically alongside word weight especially when we change word weight from 0.5 to 1.0 .

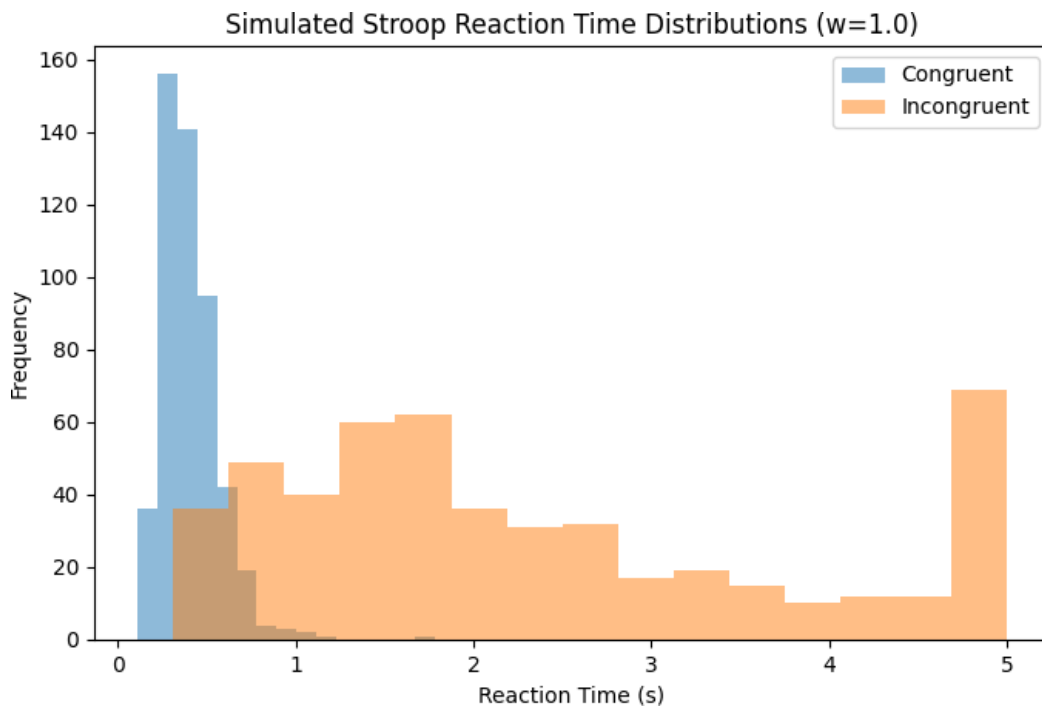


Figure 1: Reaction time for word weight = 1

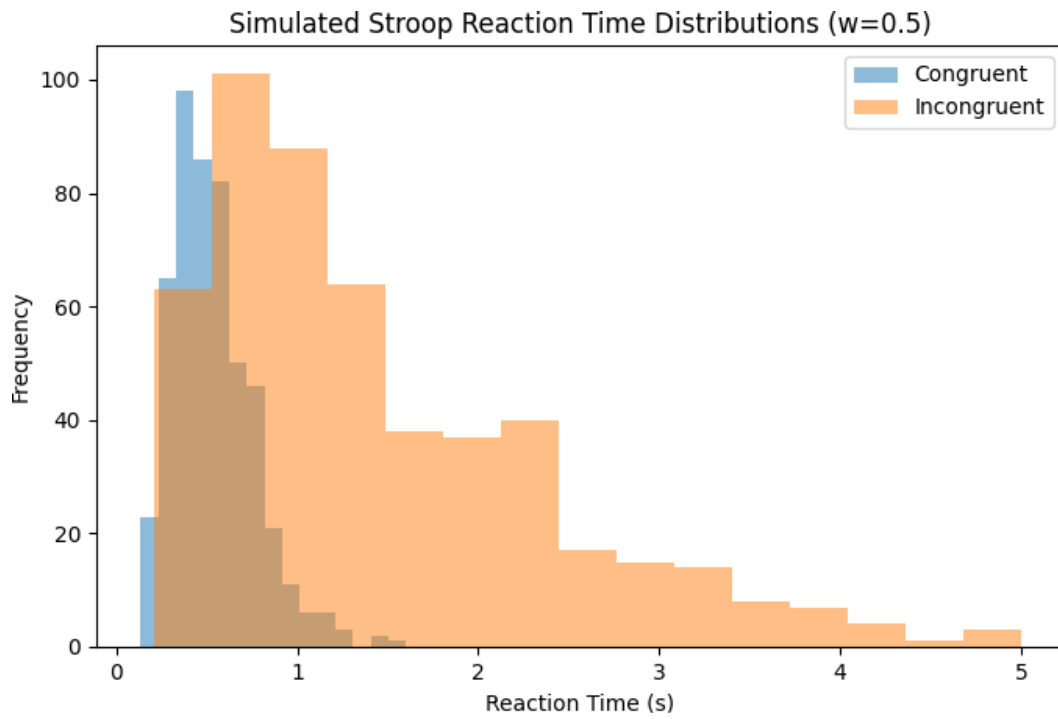


Figure 2: Reaction time for word weight = 0.5

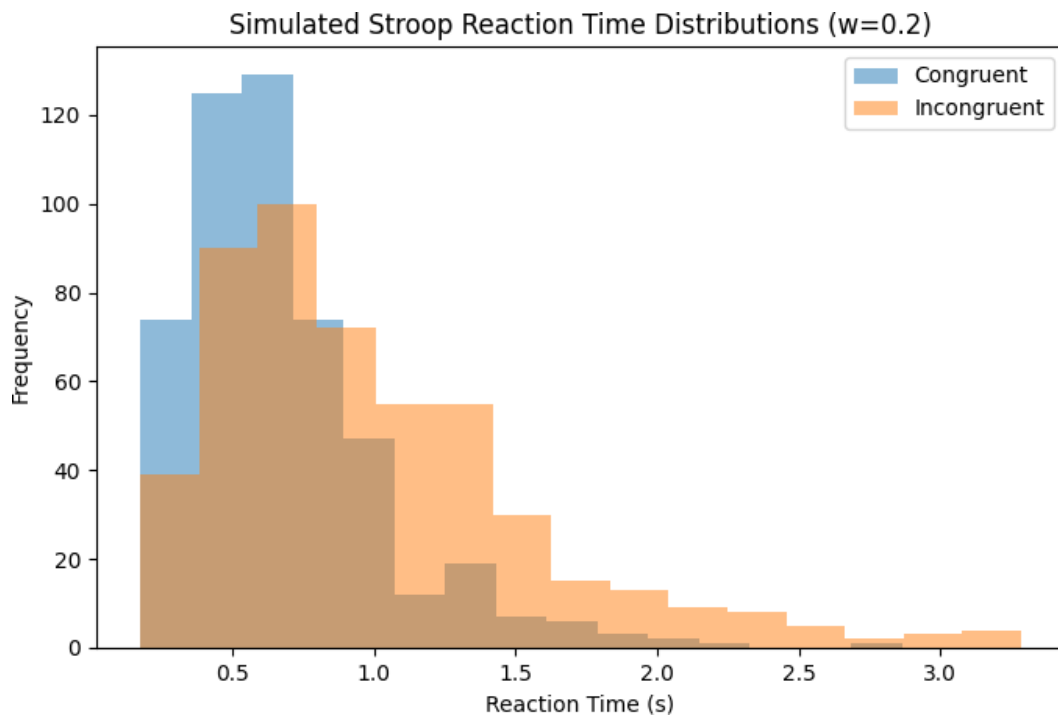


Figure 3: Reaction time for word weight = 0.2



Figure 4: Reaction time based on word weight



Figure 5: Error rate based on word weight

Discussion

The results of the simulation are consistent with the Stroop Effect. In incongruent trials reaction time is increased while accuracy is decreased. This can be attributed to the interference between automatic and controlled processing. Reading the word is an automatic process, while naming the color ink requires controlled attention. This conflict results in the decision-making process slowing down. The simulation also shows that congruent trials are more stable with lower reaction times and almost no errors.

Within the Drift Diffusion Model framework, these results can be explained by how quickly evidence builds up toward a decision. In incongruent trials, conflicting information slows down this process, so the model takes longer to reach the decision threshold. This leads to longer reaction times and more errors.

However, it is worth mentioning that this is a simplified computational model. Real human participants may show more variability due to factors such as learning or fatigue.

Conclusion

In conclusion, the experiment confirmed the Stroop effect: incongruent trials produce longer reaction times and higher error rates, while congruent trials show minimal changes. These results illustrate how conflicting information interferes with selective attention and cognitive control.

References

Stroop, J. R. (1935). *Studies of interference in serial verbal reactions*. Journal of Experimental Psychology, 18(6), 643–662.

Ratcliff, R., & McKoon, G. (2008). *The diffusion decision model: Theory and data for two-choice decision tasks*. Neural Computation, 20(4), 873–922.

MacLeod, C. M. (1991). *Half a century of research on the Stroop effect: An integrative review*. Psychological Bulletin, 109(2), 163–203.

Appendix A: Code

```

#libraries
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
#scripts
import sys
import os

# Add the scripts folder to Python path
sys.path.append(os.path.abspath('../scripts'))

# Now import your scripts
from ddm import simulate_ddm
from stroop_drift import stroop_drift

def run_condition(n_trials, color_signal, word_signal, color_weight, word_weight):
    rts = []
    errors = 0
    for _ in range(n_trials):
        drift = stroop_drift(color_signal, word_signal, color_weight,
word_weight)
        rt, decision = ddm(drift)
        rts.append(rt)
        if decision != color_signal:
            errors += 1
    return np.array(rts), errors / n_trials

def run_condition(n_trials, color_signal, word_signal, color_weight, word_weight):
    """
    Run multiple trials for one Stroop condition.

    Returns:
        rts: array of reaction times
        error_rate: proportion of incorrect decisions
    """
    rts = []
    errors = 0
    for _ in range(n_trials):
        drift = stroop_drift(color_signal, word_signal, color_weight, word_weight)
        rt, decision = simulate_ddm(drift)
        rts.append(rt)
        if decision != color_signal:
            errors += 1
    return np.array(rts), errors / n_trials

# Simulation parameters
n_trials = 500
word_weights = [1.0, 0.5, 0.2] # higher = more interference

# Store results
results = {}

for w in word_weights:
    # Congruent trial
    rt_cong, err_cong = run_condition(n_trials, color_signal=1, word_signal=1,
color_weight=1.0, word_weight=w)

```

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# Incongruent trial
rt_incong, err_incong = run_condition(n_trials, color_signal=1, word_signal=-1,
color_weight=1.0, word_weight=w)

# Save results
results[w] = {
    "rt_cong": rt_cong,
    "rt_incong": rt_incong,
    "err_cong": err_cong,
    "err_incong": err_incong
}

# Print summary
for w, res in results.items():
    print(f"Word weight: {w}")
    print(f" Congruent RT mean: {res['rt_cong'].mean():.3f}s, error rate:
{res['err_cong']:.2f}")
    print(f" Incongruent RT mean: {res['rt_incong'].mean():.3f}s, error rate:
{res['err_incong']:.2f}")

all_data = []
for w, res in results.items():
    for rt in res["rt_cong"]:
        all_data.append({"word_weight": w, "condition": "congruent", "rt": rt})
    for rt in res["rt_incong"]:
        all_data.append({"word_weight": w, "condition": "incongruent", "rt": rt})

df_all = pd.DataFrame(all_data)
df_all.to_csv('../data/stroop_all_results.csv', index=False)
print("Simulation results saved to data/stroop_all_results.csv")

for w, res in results.items():
    plt.figure(figsize=(8,5))

    # Plot congruent and incongruent RTs
    plt.hist(res['rt_cong'], bins=15, alpha=0.5, label='Congruent')
    plt.hist(res['rt_incong'], bins=15, alpha=0.5, label='Incongruent')

    plt.xlabel("Reaction Time (s)")
    plt.ylabel("Frequency")
    plt.title(f"Simulated Stroop Reaction Time Distributions (w={w})")
    plt.legend()

    # Save figure with w in filename
    plt.savefig(f'../figures/rt_distributions_w{w}.png')
    plt.show()
    print(f"Plot saved to figures/rt_distributions_w{w}.png")

# Extract mean RTs and error rates
word_weights = list(results.keys())
rt_cong_means = []
rt_incong_means = []

for w in word_weights:
    res = results[w]
    rt_cong_means.append(res['rt_cong'].mean()) # mean of 500 trials
    rt_incong_means.append(res['rt_incong'].mean())

```



```

err_cong = []
err_incong = []
for w in word_weights:
    res = results[w]
    err_cong.append(res['rt_cong'].mean())    # mean of 500 trials
    err_incong.append(res['rt_incong'].mean())
err_cong = [results[w]['err_cong'] for w in word_weights]
err_incong = [results[w]['err_incong'] for w in word_weights]

plt.figure(figsize=(8, 5))
plt.plot(word_weights, rt_cong_means, marker='o', label='Congruent')
plt.plot(word_weights, rt_incong_means, marker='o', label='Incongruent')
plt.xlabel('Word Weight')
plt.ylabel('Mean Reaction Time (s)')
plt.title('Mean Reaction Time vs Word Weight')
#plt.gca().invert_xaxis() # optional
plt.legend()
plt.grid(True)
plt.savefig('../figures/Mean_Reaction_VS_Word_Weight.png')
plt.show()
print("Plot saved to figures/Mean_Reaction_VS_Word_Weight.png")

```